

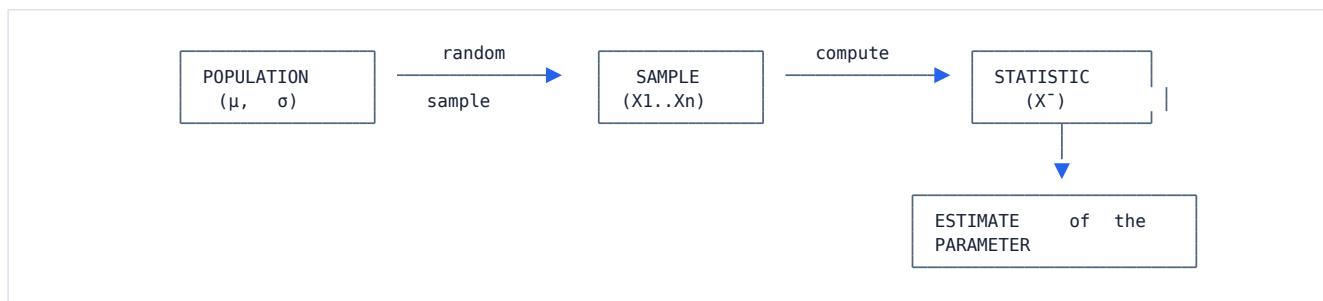
09 THEORY OF ESTIMATION (ppt5, 26 pages, lectures 19-21)

1. The vocabulary (p003-p004)

PARAMETER	a numerical characteristic of a POPULATION
	μ, σ^2, p
STATISTIC	a number computed FROM SAMPLE DATA
	$\bar{X}, S^2, \hat{p}$
ESTIMATOR	a FUNCTION of the sample used to estimate
	a parameter. It is a RANDOM VARIABLE.
	e.g. $\bar{X}$ as an estimator of μ
ESTIMATE	the NUMBER you get once data is plugged in
	e.g. $\bar{x} = 52$
PARAMETER SPACE	all possible values of the parameter
	$\Theta = (-\infty, \infty)$ for μ

ESTIMATOR vs ESTIMATE - one mark, easy to lose
 $\bar{X}$ = the RULE (an estimator, a random variable)
 $\bar{x} = 205$ = the NUMBER (the estimate, once data is in)

2. The estimation pipeline (p004)



3. Point vs interval estimation (p005-p007)

POINT ESTIMATION	INTERVAL ESTIMATION
a SINGLE value $\mu \approx \bar{X}$	a RANGE of values with a stated confidence level
✓ simple, easy to report x does not show uncertainty	✓ accounts for uncertainty ✓ more informative

the picture from p007:

POINT

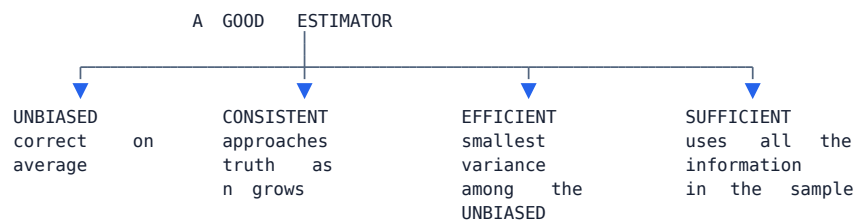
210 ms
"one number"

INTERVAL

204ms 216ms
"a plausible range"

CI = Point Estimate $\pm$ Margin of Error

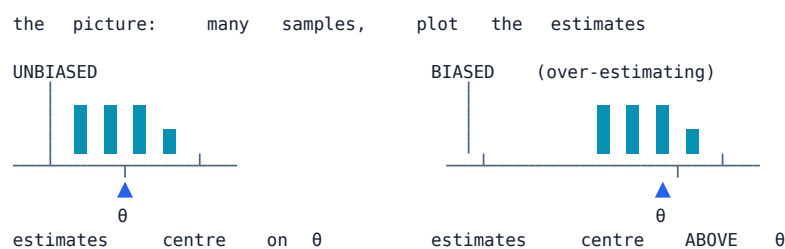
4. The four characteristics of a good estimator (p008)



"good = little systematic error + stability + precision"

5. Unbiasedness (p009)

$\hat{\theta}$ is UNBIASED for θ if $E(\hat{\theta}) = \theta$
 $\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta$



"CORRECT ON AVERAGE, NOT PER SAMPLE"

An unbiased estimator can be badly wrong on any ONE sample.
 Unbiasedness is a statement about the LONG RUN.

6. Consistency (p010-p012)

$\hat{\theta}_n$ converges in probability to θ as $n \rightarrow \infty$
 $\hat{\theta}_n \xrightarrow{P} \theta$
 i.e. for any $\epsilon > 0$: $P(|\hat{\theta}_n - \theta| < \epsilon) \rightarrow 1$ as $n \rightarrow \infty$

SUFFICIENT CONDITIONS FOR CONSISTENCY (a sequence T_n):

1. $E(T_n) \rightarrow \theta$ as $n \rightarrow \infty$
 2. $\text{Var}(T_n) \rightarrow 0$ as $n \rightarrow \infty$
- Both hold $\Rightarrow T_n$ is consistent.

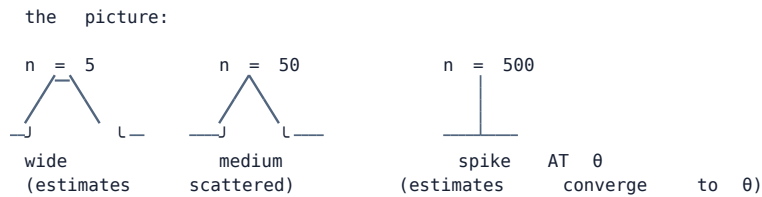
WHY X IS CONSISTENT (p012) - the standard answer:

CONDITION 1 | CONDITION 2 | CONCLUSION

$E(X) = \mu$ | $\text{Var}(X) = \sigma^2/n$ | both hold

so $E(X) \rightarrow \mu$ | so $\sigma^2/n \rightarrow 0$ | $\Rightarrow X$ is consistent

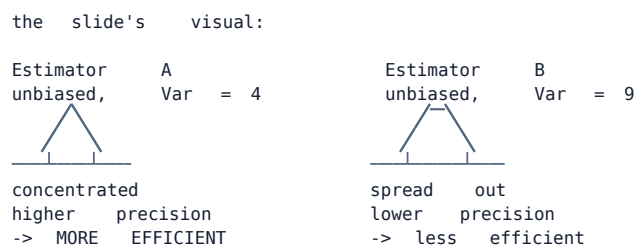
"more observations $\rightarrow$ smaller variance $\rightarrow$ more stable estimate"



7. Efficiency (p013-p014)

θ_1 and θ_2 both UNBIASED for θ :
 if $\text{Var}(\theta_1) < \text{Var}(\theta_2)$ then θ_1 is MORE EFFICIENT

Among all UNBIASED estimators, the most efficient has the SMALLEST variance.
 Efficiency $\propto 1/\text{Variance}$



THE ORDER OF OPERATIONS IN ANY COMPARISON:
 STEP 1 find which are UNBIASED (compute E of each)
 STEP 2 a biased one is ELIMINATED, even if its variance is smaller ("efficiency" only compares the unbiased)
 STEP 3 compare variances of the SURVIVORS, smallest wins

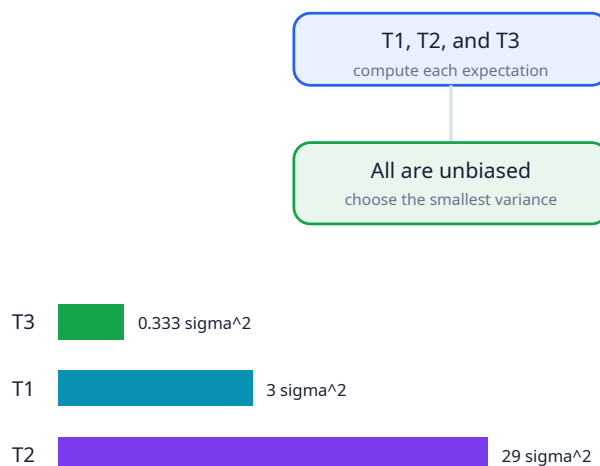
8. WORKED COMPARISON SET 1 (p015-p018) - the flagship question

Sample of size 5 from $N(\mu, \sigma^2)$.	$(X_1 \dots X_5)$
$t_1 = (X_1 + X_2 + X_3 + X_4 + X_5)/5$	<- the sample mean
$t_2 = (X_1 + X_2)/2 + X_3$	
$t_3 = (2X_1 + X_2 + \lambda X_3)/3$	<- λ chosen for unbiasedness
STEP 1: FIND λ $E(t_3) = [2E(X_1) + E(X_2) + \lambda E(X_3)]/3$ $= (2\mu + \mu + \lambda\mu)/3 = [(3+\lambda)/3] \mu$ for unbiasedness: $(3+\lambda)/3 = 1 \rightarrow \lambda = 0$ so $t_3 = (2X_1 + X_2)/3$	
STEP 2: WHICH ARE UNBIASED	
$E(t_1) = (\mu + \mu + \mu + \mu + \mu)/5 = \mu$	✓ unbiased
$E(t_2) = (\mu + \mu)/2 + \mu = 2\mu$	✗ BIASED
$E(t_3) = (2\mu + \mu)/3 = \mu$	✓ unbiased
STEP 3: COMPARE THE VARIANCES (of the unbiased ones)	
$\text{Var}(t_1) = \sigma^2/5$	$= 0.2000 \sigma^2$

STEP 3: COMPARE THE VARIANCES (of the unbiased ones)					
$\text{Var}(t_3) = (4\sigma^2 + \sigma^2)/9$			$= 5\sigma^2/9 = 0.5556 \sigma^2$		
ordering:	$\sigma^2/5$	<	$5\sigma^2/9$	<	$3\sigma^2/2$
(t1)	(t3)	(t2, biased)			
BEST = t1 = $\bar{X}$, the sample mean					
the variance bar chart: t1 0.200 σ^2 <- WINNER t3 0.556 σ^2 t2 1.500 σ^2 (biased, disqualified)					

9. WORKED COMPARISON SET 2 (p019-p021) - with consistency

Estimator set 2: all unbiased, compare variance



Consistency needs a sequence T_n as n grows; fixed $n=3$ cannot prove it.

Unbiased estimators compared by variance

Sample of size 3. T_1, T_2, T_3 estimate μ .

$$T_1 = X_1 + X_2 - X_3$$

$$T_2 = 2X_1 + 3X_3 - 4X_2$$

$$T_3 = (\lambda X_1 + X_2 + X_3)/3$$

(i) ARE T_1 and T_2 UNBIASED?

$$E(T_1) = \mu + \mu - \mu = \mu$$

✓

$$E(T_2) = 2\mu + 3\mu - 4\mu = \mu$$

✓

BOTH unbiased

(ii) FIND λ FOR T_3

$$E(T_3) = (\lambda\mu + \mu + \mu)/3 = ((\lambda+2)/3)\mu$$

$$((\lambda+2)/3)\mu = \mu \rightarrow \lambda = 1$$

$$\text{so } T_3 = (X_1 + X_2 + X_3)/3 = \bar{X}, \text{ the sample mean}$$

(iii) IS T3 CONSISTENT?

With $\lambda=1$, this fixed-n statistic is the sample mean for $n=3$.

That fact alone does not establish consistency. The sequence $\bar{X}_n$ is consistent under the usual iid assumptions because $\text{Var}(\bar{X}_n) = \sigma^2/n \rightarrow 0$.

(iv) VARIANCES

$$\text{Var}(T1) = \text{Var}(X1) + \text{Var}(X2) + \text{Var}(X3) = 3\sigma^2$$

$$\text{Var}(T2) = 4\text{Var}(X1) + 9\text{Var}(X3) + 16\text{Var}(X2) = 29\sigma^2$$

$$\text{Var}(T3) = (1/9)(3\sigma^2) = \sigma^2/3$$

minimum $\rightarrow$ T3 is the BEST estimator

the variance bar chart:



THE CONSISTENCY TRAP (worth memorising):

T1's variance is $3\sigma^2$ for a FIXED $n=3$.

That is NOT a consistency statement.

Consistency is about behaviour AS n GROWS.

A fixed-n variance, however small or large, says nothing about consistency.

10. Sufficiency (the lms-theory extra)

an estimator is SUFFICIENT if it uses ALL the information

about θ present in the sample

the test is the NEYMAN-FISHER factorisation

examples in the corpus: the Poisson sample sum, the exponential sample mean, the sufficient statistics MCQ on MTE 2025-26 Q3

11. WORKED Real-Life 1: response time (p019)

response times of 8 requests: 180,210,195,220,205,190,200,240

$$\bar{x} = (\text{sum})/8 = 1640/8 = 205 \text{ ms}$$

ESTIMATOR = $\bar{X}$

(the rule)

ESTIMATE

= 205 ms

(the number)

THE QUESTION ASKS FOR BOTH "the estimator and the estimate". Two marks. Do not give only the number.

12. WORKED Real-Life 2: packet success (p020)

500 packets, 465 delivered. Estimate p .

$$\hat{p} = X/n = 465/500 = 0.93 = 93\%$$

estimator used: $\hat{p}$

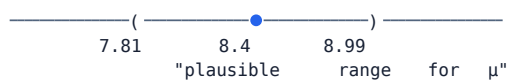
13. WORKED Real-Life 3: battery life (p021) - the CI boundary item

$n = 25$, $\bar{x} = 8.4$ hours, $\sigma = 1.5$ known. 95% CI, $z(0.975) = 1.96$

This z interval assumes a normal population or a separately justified normal approximation. A sample size of 25 alone does not supply that justification.

$$\begin{aligned} \text{CI} &= \bar{x} \pm z(\sigma/\sqrt{n}) \\ &= 8.4 \pm 1.96(1.5/5) \\ &= 8.4 \pm 1.96(0.3) \\ &= 8.4 \pm 0.588 \\ &= (7.81, 8.99) \text{ hours} \end{aligned}$$

the picture:



NOTE: the CI MECHANICS belong to lecture 25 (BOUNDARY for this audit), but "point and interval estimation" IS named in the MTE syllabus line 15. Keep the formula in reserve; do not drill the mechanics.

14. The comparison quick table (p022)

#	PARAMETER	STATISTIC	RESULT
1	μ	$\bar{X}$	205 ms
2	p	$\hat{p}$	0.93
3	μ	$\bar{X}$	95% CI (7.81,8.99)

15. The standard unbiased results (the toolkit)

ESTIMATOR	ESTIMATES	VARIANCE
$\bar{X}$	μ	σ^2/n
$S^2 = \sum(X_i - \bar{x})^2/(n-1)$	σ^2	(the n-1 matters)
$\hat{p} = X/n$	p	$p(1-p)/n$

WHY n-1 AND NOT n IN S^2

The deviations are measured from $\bar{x}$, which ALREADY used the data. Using n would make S^2 too small on average. Dividing by n-1 corrects that. It is the price of estimating the mean from the same data.

16. MSE, the honest comparison (formula sheet addition)

$$\text{MSE}(\theta) = \text{Var}(\theta) + [\text{Bias}(\theta)]^2$$

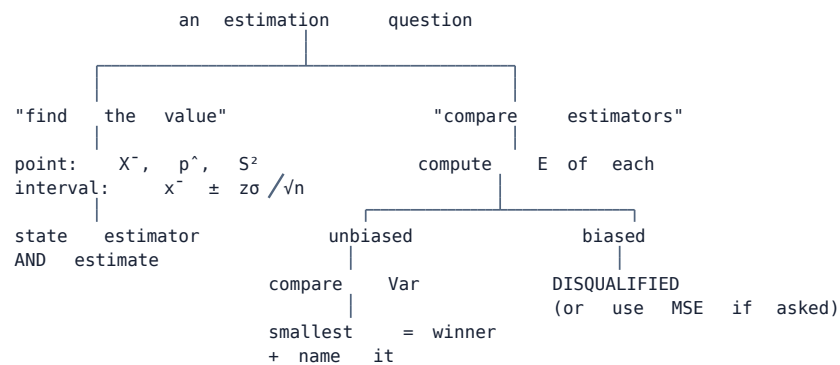
the picture:

$$\text{MSE} = \underbrace{\text{VARIANCE}}_{\text{spread}} + \underbrace{\text{BIAS}^2}_{\text{off-target}}$$



Efficiency ONLY compares unbiased estimators.
 MSE lets a biased estimator compete, by paying for its
 bias in the second term. The exam could introduce this.

THE ESTIMATION DECISION TREE (for any question in this block)



That is the theory of estimation. The estimator-comparison layout (unbiased -> variances -> name the winner) is the 6-to-8 mark question of this block.